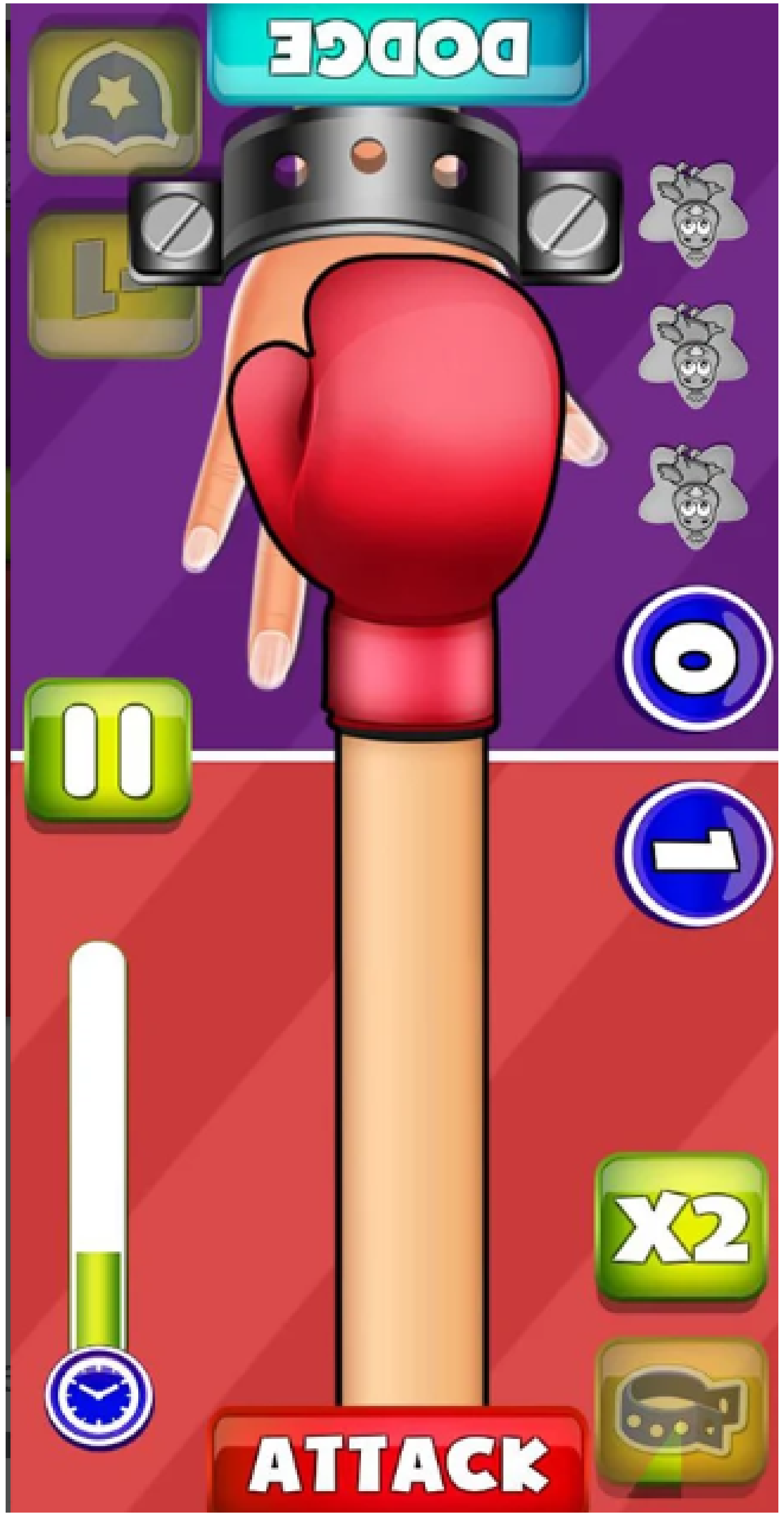
HAND SLAP

How the Idea Came

We were exploring traditional games like tug of war, mastermind, etc.

One of these games was Hand Slap



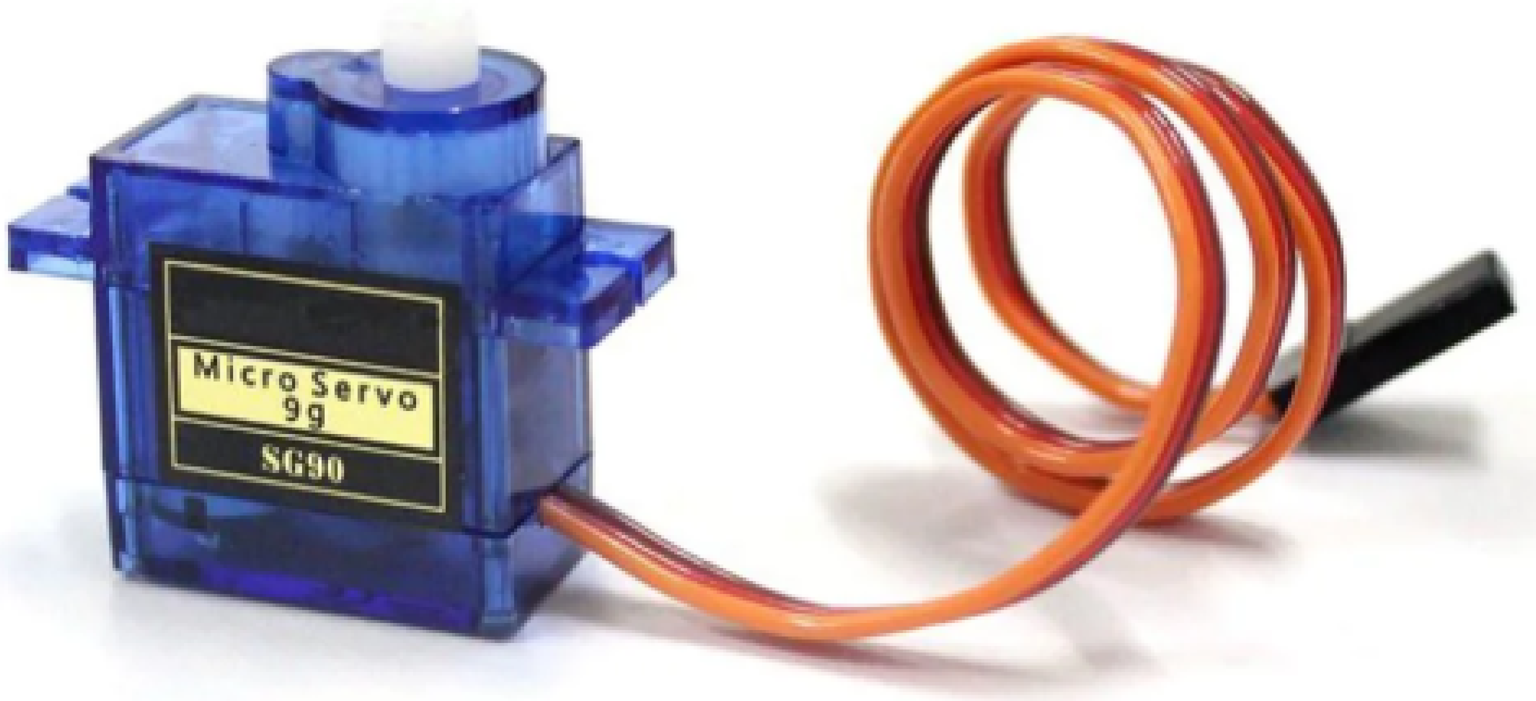
Technical Feasibility

In traditional hand slap, the attacker player's hand goes over the defender player's hand, while the defender player's job is to dodge it.

We translated the main aim of the game, and instead of making one hand hover over the other, we made it "chase"

Components We Used

Since the assignment was to make the game interactive, we used 2 push buttons - assigning one for the attacker and one for the dodger.



To make the hands do the sideways motion, we decided to use 2 servo motors, so they move in the exact range of motion we wanted (0-120°)

Logic Used

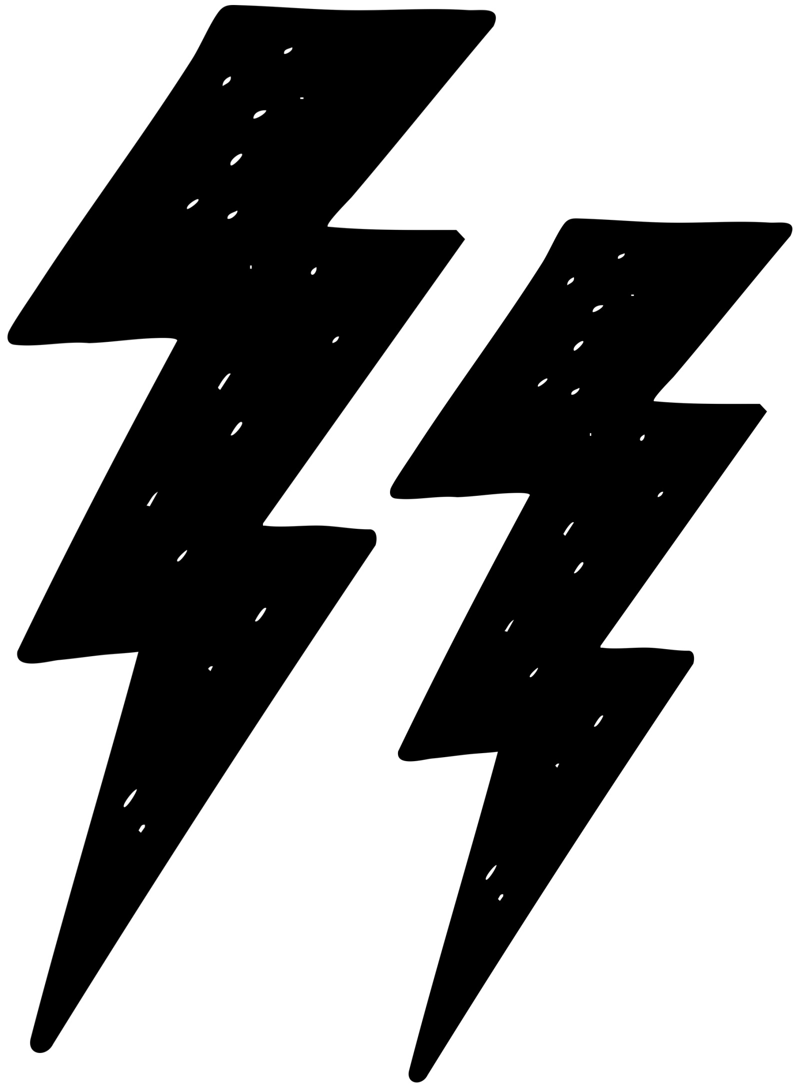
INPUT

OUTPUT

The value of the buttons needs to be continuously read, so that when the button is pressed, the respective player's servo motor moves.

We've done this by putting the "button.value()" under the "while True" loop and then writing "if" loops that correspond to the pressing of the button, which triggers the servo motor.

Pain Points

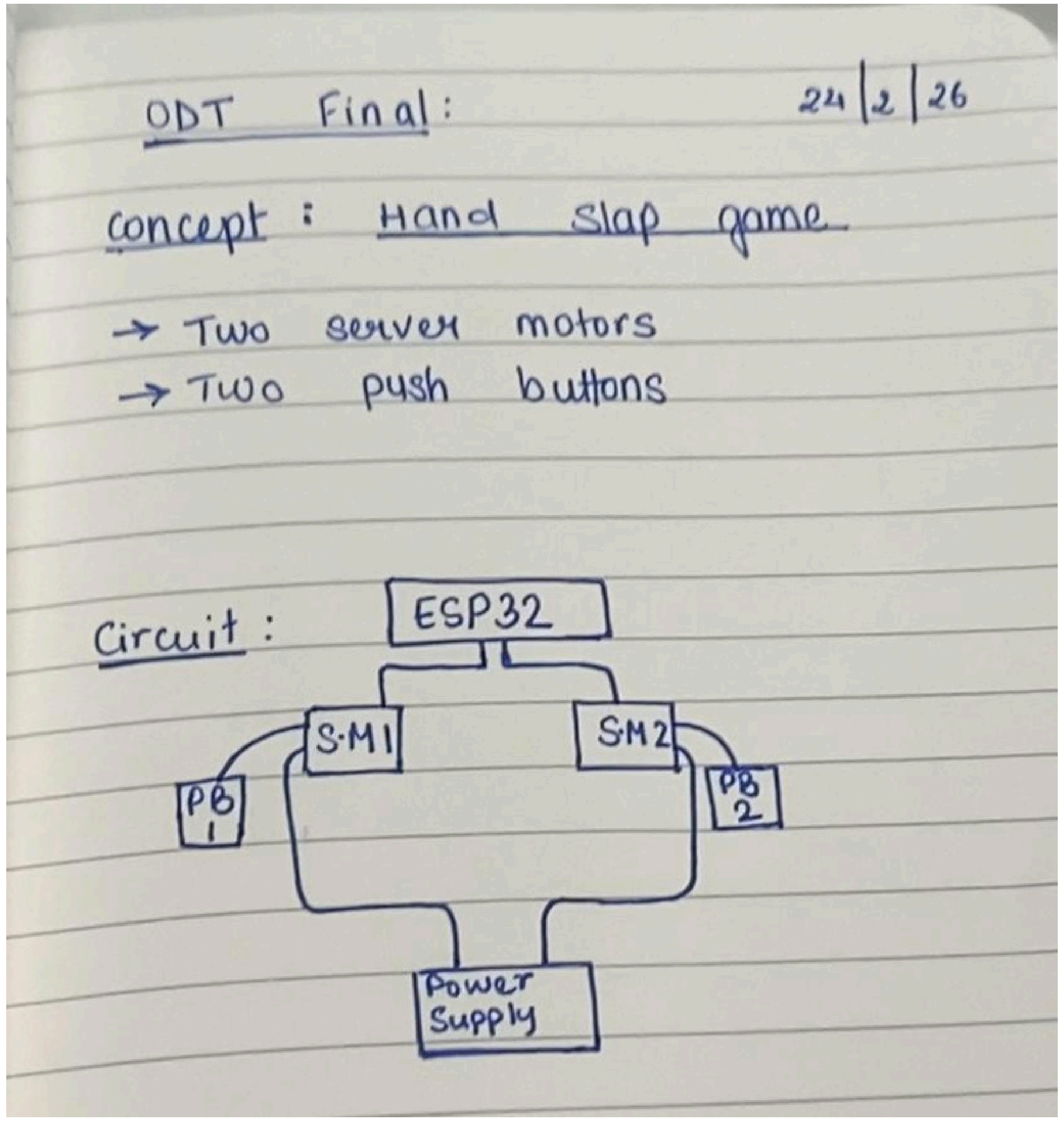


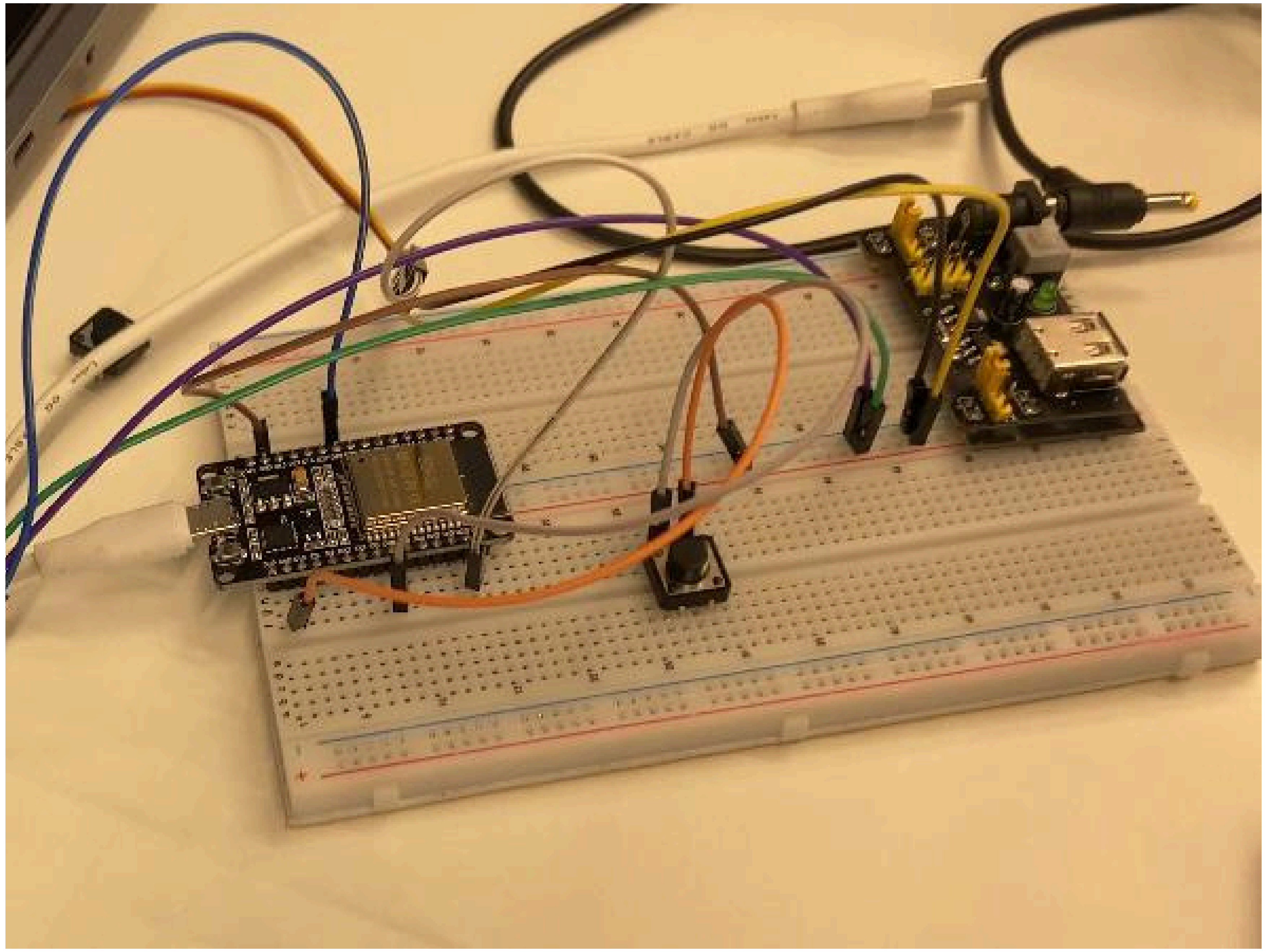
- Time delay between movement of both the hands
- How to cover all the wires and accomodating for the height of the push buttons
- Human errors while testing push buttons - wanted to see what woul happen if both players pressed the button at the same time

Individual Contribution (Dhriti)

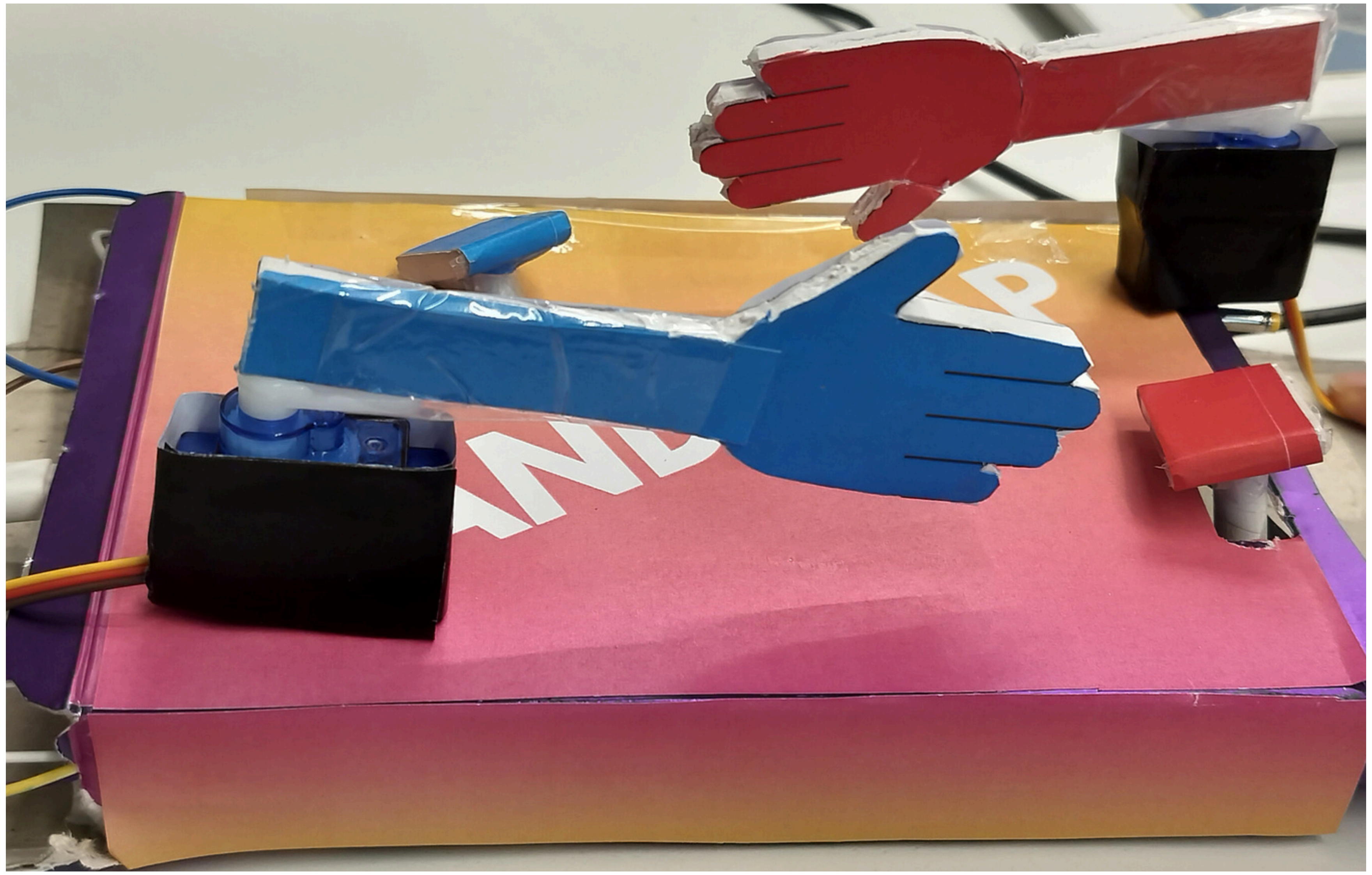
- Ideation
- Logic making
- Circuit attachment
- Coding
- Adding attachments
- Presentation Making

Circuit


ODT Final:
24|2|26
concept : Hand slap game
→ Two server motors
→ Two push buttons
Circuit :
ESP32
S·M1
SM2
PB 1
PB 2
Power Supply



Final Outcome



Learnings

- Start with writing the logic of the flow of the system and code each component individually first.
- How to change the direction of the servo motor.
- The time difference it takes between two “if” loops – so we had to experiment with the time delay so there isn’t too much difference between both the player’s hand movements
- Even though the idea may seem easy, execution reveals errors.

Code

```
from machine import Pin, PWM
import time

servo1 = PWM(Pin(22))
servo1.freq(50)
servo2 = PWM(Pin(26))
servo2.freq(50)
button1 = Pin(18,Pin.IN,Pin.PULL_UP)
button2 = Pin(33,Pin.IN,Pin.PULL_UP)
while True:
    value1 = button1.value()
    value2 = button2.value()

    #attack
    if value1 == 0 :
        servo2.duty(90)
        time.sleep(0.1)
        servo2.duty(45)


    #defence
    if value2 == 0 :
        servo1.duty(45)
        time.sleep(0.1)
        servo1.duty(90)
    time.sleep(0.2)
```

Thank you!